

Project Design Phase-II

Technology Stack (Architecture & Stack)

Date: 18 June 2026

Project Name: Online Complaint Registration System

Technical Architecture

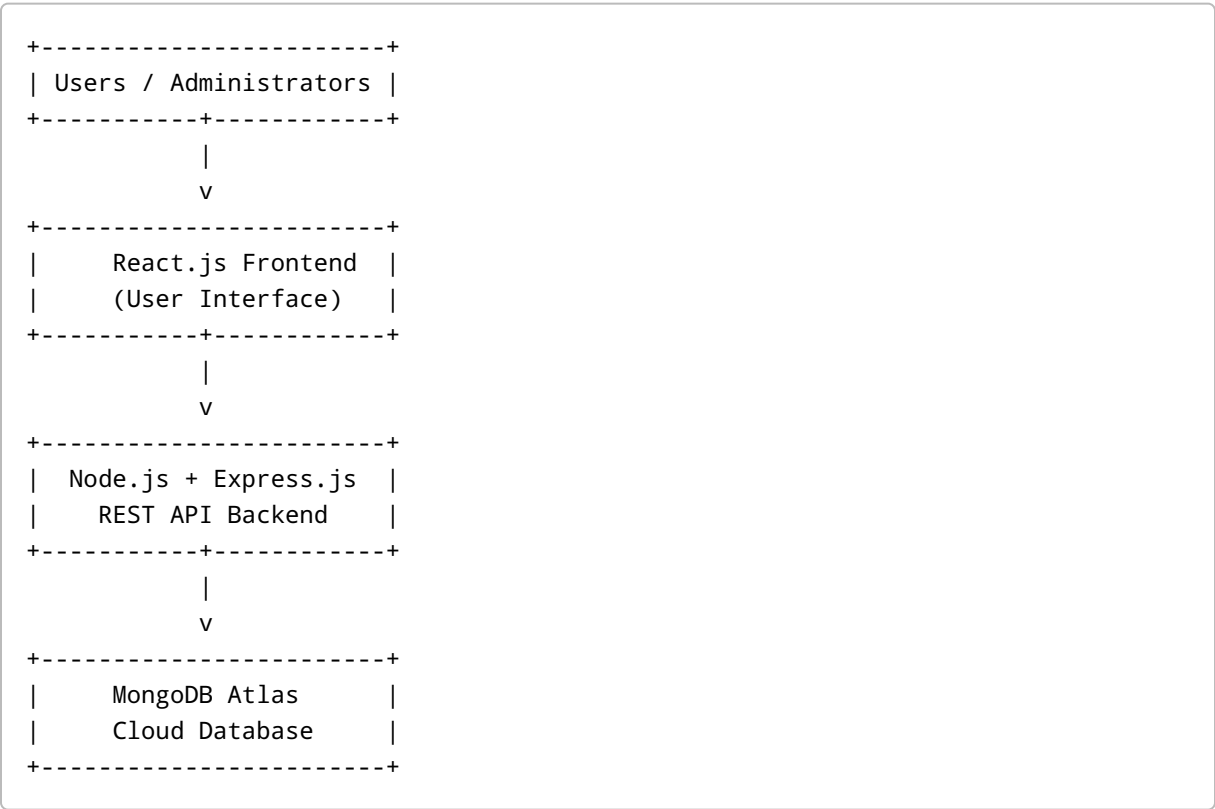


Table-1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Web application interface for users and administrators	React.js, HTML5, CSS3, JavaScript

S.No	Component	Description	Technology
2	Application Logic-1	User authentication and authorization	Node.js, Express.js
3	Application Logic-2	Complaint registration and complaint tracking	Node.js, Express.js
4	Application Logic-3	Complaint management and notification workflow	Node.js, Express.js
5	Database	Stores users, complaints, status updates, and notifications	MongoDB
6	Cloud Database	Cloud-hosted database service	MongoDB Atlas
7	File Storage	Stores application assets and uploaded documents	Local Storage / Cloud Storage
8	External API-1	REST API communication between frontend and backend	Axios API Calls
9	External API-2	Authentication token verification	JWT
10	Machine Learning Model	Not applicable in current version	N/A
11	Infrastructure (Cloud)	Application deployment and hosting	Vercel (Frontend), Render (Backend)

Table-2: Application Characteristics

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	Frameworks used for application development	React.js, Node.js, Express.js
2	Security Implementations	Password encryption, authentication, protected routes	JWT, bcryptjs
3	Scalable Architecture	Modular MERN architecture supporting future expansion	MERN Stack
4	Availability	Cloud deployment ensures access from anywhere	Vercel, Render, MongoDB Atlas
5	Performance	Fast API communication and optimized frontend rendering	React.js, Express.js, Axios

Deployment Details

Frontend Deployment

<https://vip-c2-online-complaint-registerati.vercel.app/>

Backend Deployment

<https://complaint-registration.onrender.com>

Database

MongoDB Atlas

Architecture Benefits

- Secure user authentication and authorization.
 - Centralized complaint management.
 - Real-time complaint tracking and status updates.
 - Cloud-based database access.
 - Fast and responsive user experience.
 - Easy maintenance and scalability.
 - Efficient communication between users and administrators.
 - Supports future enhancements such as AI-based complaint categorization and analytics dashboards.
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Technology Advantages

React.js

- Component-based architecture
- Fast rendering using Virtual DOM
- Responsive user interface

Node.js & Express.js

- Efficient backend processing
- RESTful API development
- Scalable server architecture

MongoDB Atlas

- Flexible NoSQL database
- Cloud-hosted infrastructure

- High availability and scalability

JWT Authentication

- Secure user sessions
 - Token-based authentication
 - Improved application security
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Conclusion

The Online Complaint Registration System is built using a modern MERN Stack architecture that provides scalability, security, reliability, and high performance. The selected technology stack ensures efficient complaint registration, complaint tracking, complaint management, and complaint resolution while supporting future enhancements and system growth.

The architecture enables secure communication between users, administrators, and the database, making the platform robust, maintainable, and suitable for real-world deployment.